

# Assignment Report: Git + GitHub Guestbook Exercise

Course: Cloud Computing and Data Engineering

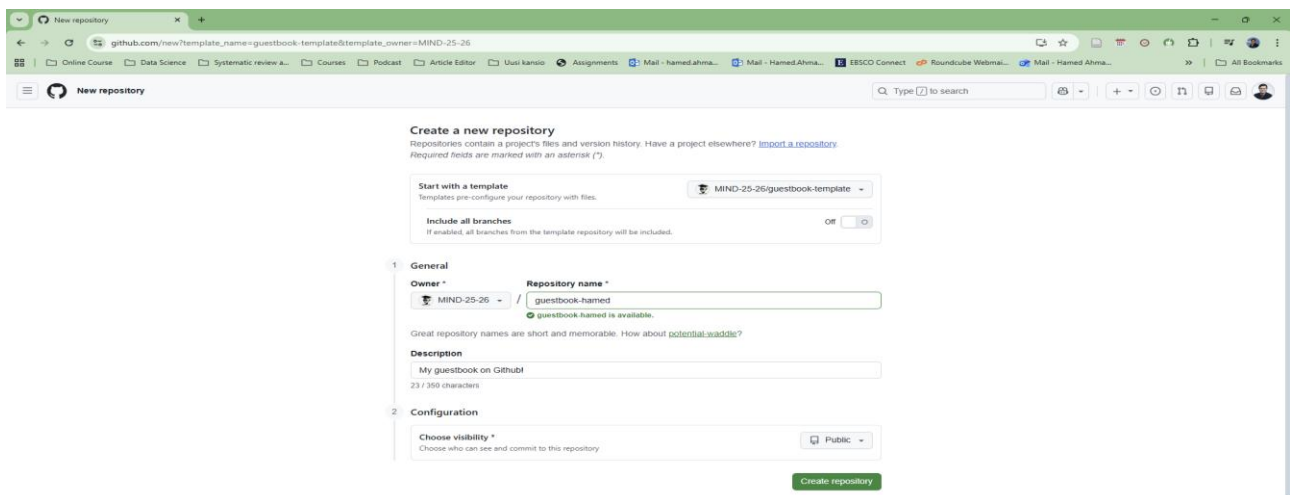
By: Hamed Ahmadinia

## Introduction

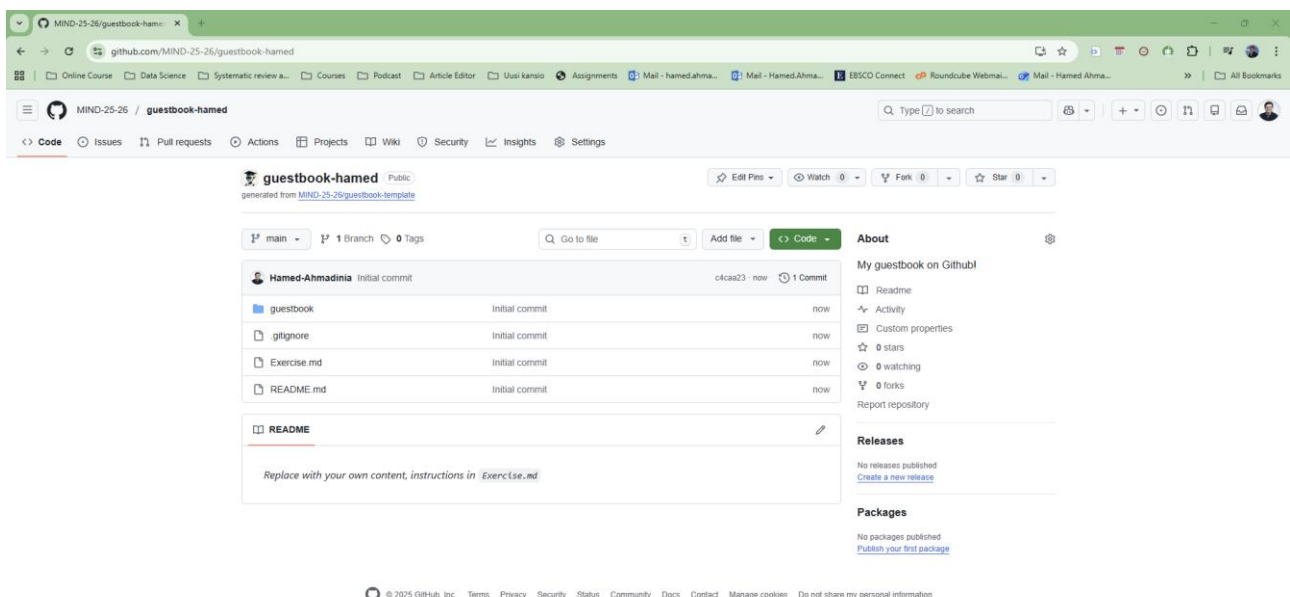
This report documents the steps taken to complete the GitHub guestbook assignment, which involved creating a personal repository from a template, working locally with Git and Visual Studio Code, and collaborating through forks and pull requests. The assignment was divided into two parts: preparing my own guestbook repository and contributing to another student's guestbook. Screenshots are provided to demonstrate each key step and verify successful completion. The report concludes with reflections on challenges encountered and lessons learned.

## Part 1 – Prepare your own guestbook

*Step 1: Creation of a new repository using the provided guestbook-template within the MIND organization, with the repository named guestbook-hamed and set to public visibility.*



*Step 2: Overview of the newly created guestbook-hamed repository on GitHub, confirming that it was successfully generated from the guestbook-template within the MIND organization and is publicly accessible.*

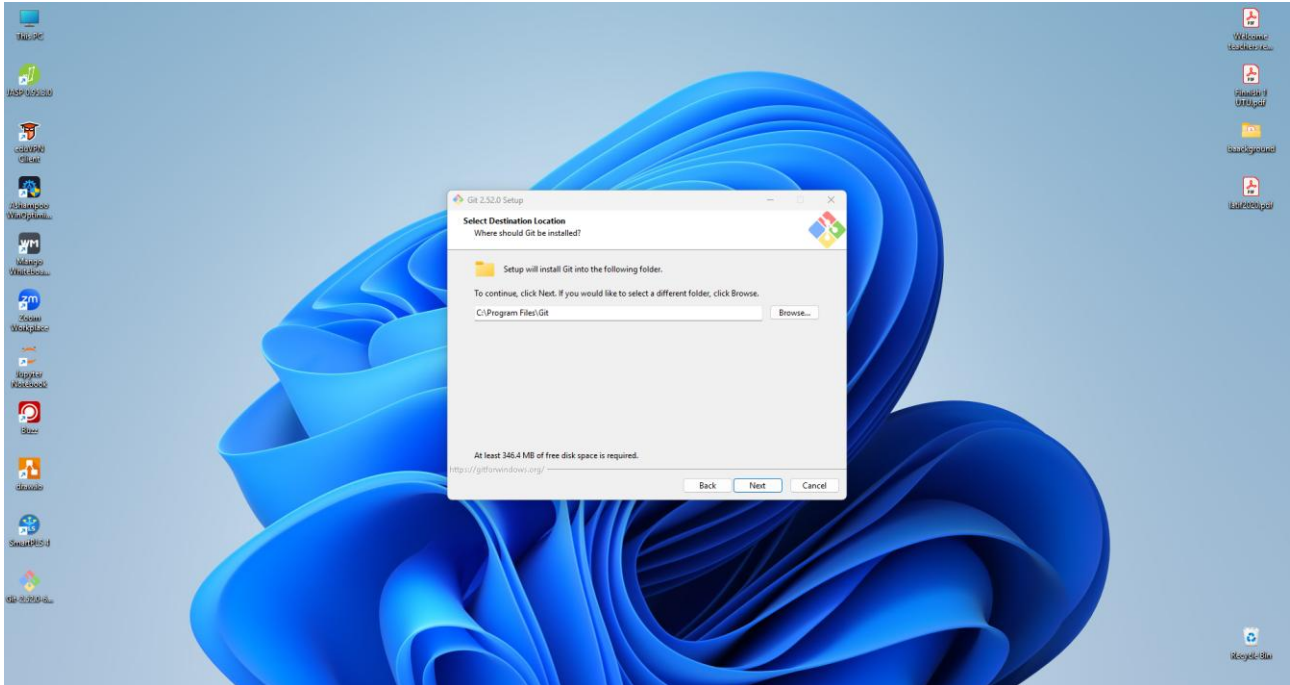


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*Step 3: As I Did not have Git for Windows installed in my laptop, I installed it in order to ensure that the Git command-line tools are available on the local machine for cloning repositories and managing version control operations.*



*Step 4: Verification and troubleshooting of SSH authentication in Windows PowerShell, showing initial permission errors followed by successful SSH authentication with GitHub after explicitly specifying the correct SSH key.*

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

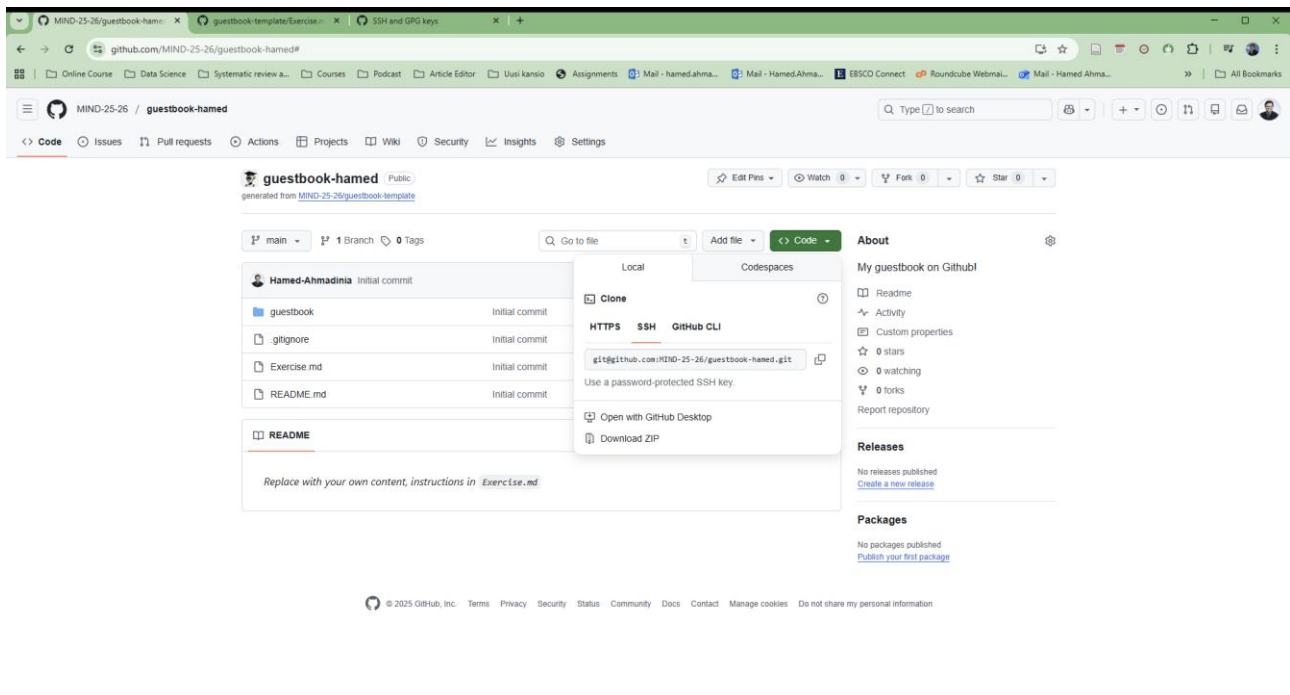
PS C:\WINDOWS\System32> cat ~/.ssh/pouta_key.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIGDGbRxBHntbnNEaIcL+ObRUin2wnNiBrNonL/LODrnV hamed@Hamed
PS C:\WINDOWS\System32> ssh -T git@github.com
The authenticity of host 'github.com (140.82.121.4)' can't be established.
ED25519 key fingerprint is SHA256:+DiY3wvvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvC0qU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
git@github.com: Permission denied (publickey).
PS C:\WINDOWS\System32> cat ~/.ssh/pouta_key.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIGDGbRxBHntbnNEaIcL+ObRUin2wnNiBrNonL/LODrnV hamed@Hamed
PS C:\WINDOWS\System32> ^C
PS C:\WINDOWS\System32> ssh -T git@github.com
git@github.com: Permission denied (publickey).
PS C:\WINDOWS\System32> ssh -i ~/.ssh/pouta_key -T git@github.com
Hi Hamed-Ahmadinia! You've successfully authenticated, but GitHub does not provide shell access.
PS C:\WINDOWS\System32>
```

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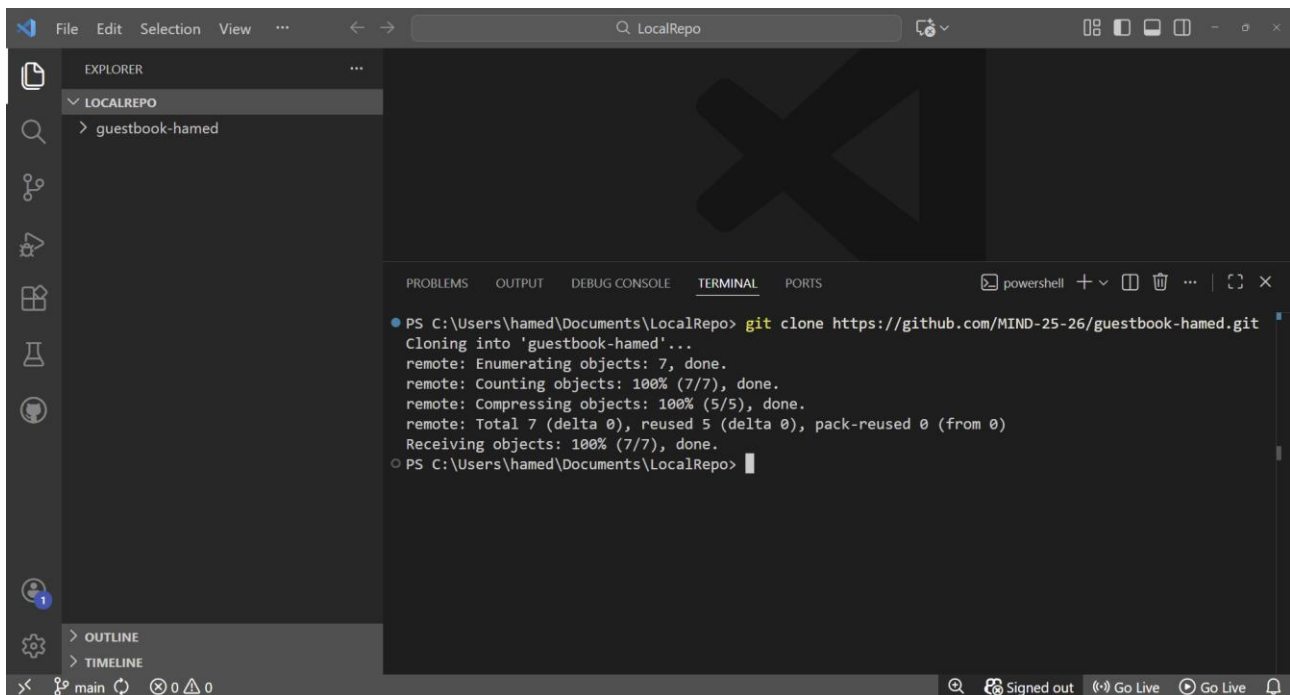
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*Step 5: I accessed the repository's **Code** menu on GitHub to copy the SSH clone URL, which is required for cloning the repository locally using SSH authentication.*



*Step 6: I successfully cloned the guestbook-hamed repository into my local working directory using the GitHub HTTPS URL in the VS Code integrated terminal.*

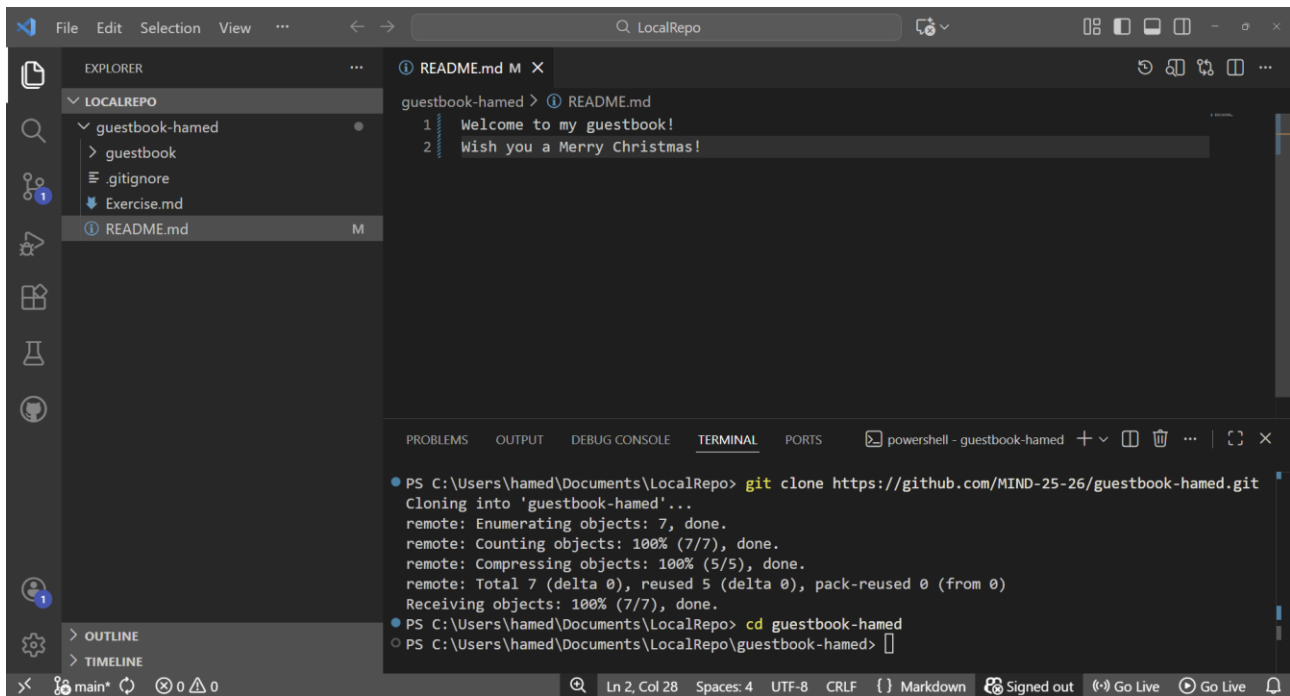


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### Step 7:



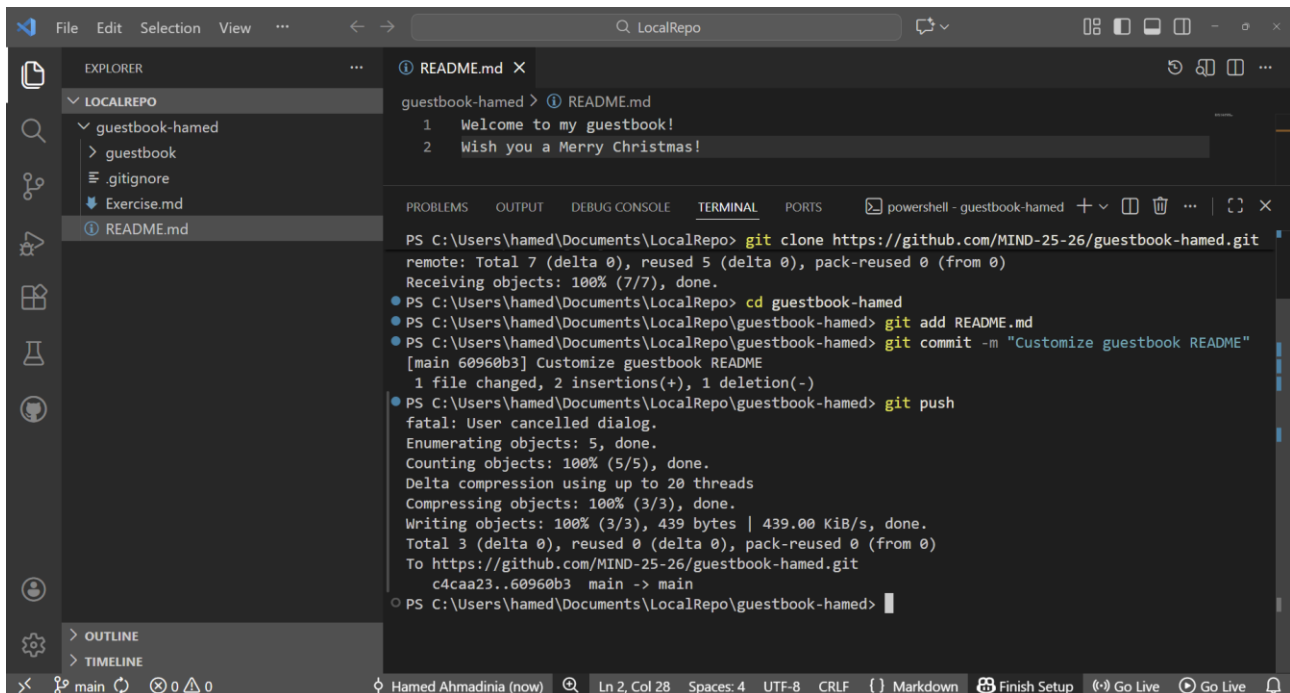
The screenshot shows the Visual Studio Code interface. The Explorer panel on the left shows a project named 'LOCALREPO' with a subdirectory 'guestbook-hamed' containing 'guestbook', '.gitignore', and 'Exercise.md'. The 'README.md' file is selected and open in the editor. The terminal at the bottom shows the following commands and output:

```
PS C:\Users\hamed\Documents\LocalRepo> git clone https://github.com/MIND-25-26/guestbook-hamed.git
Cloning into 'guestbook-hamed'...
remote: Enumerating objects: 7, done.
remote: Counting objects: 100% (7/7), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 7 (delta 0), reused 5 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (7/7), done.
PS C:\Users\hamed\Documents\LocalRepo> cd guestbook-hamed
PS C:\Users\hamed\Documents\LocalRepo\guestbook-hamed>
```

The editor shows the content of 'README.md':

```
1 Welcome to my guestbook!
2 Wish you a Merry Christmas!
```

Step 8: I opened the cloned guestbook-hamed repository in VS Code, edited the README.md file by replacing the template text with my own welcome message, and navigated into the repository directory using the terminal.



The screenshot shows the Visual Studio Code interface. The Explorer panel on the left shows the 'guestbook-hamed' directory. The terminal at the bottom shows the following commands and output:

```
PS C:\Users\hamed\Documents\LocalRepo> git clone https://github.com/MIND-25-26/guestbook-hamed.git
remote: Total 7 (delta 0), reused 5 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (7/7), done.
PS C:\Users\hamed\Documents\LocalRepo> cd guestbook-hamed
PS C:\Users\hamed\Documents\LocalRepo\guestbook-hamed> git add README.md
PS C:\Users\hamed\Documents\LocalRepo\guestbook-hamed> git commit -m "Customize guestbook README"
[main 60960b3] Customize guestbook README
1 file changed, 2 insertions(+), 1 deletion(-)
PS C:\Users\hamed\Documents\LocalRepo\guestbook-hamed> git push
fatal: User cancelled dialog.
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 20 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 439 bytes | 439.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/MIND-25-26/guestbook-hamed.git
c4caa23..60960b3 main -> main
PS C:\Users\hamed\Documents\LocalRepo\guestbook-hamed>
```

The editor shows the content of 'README.md':

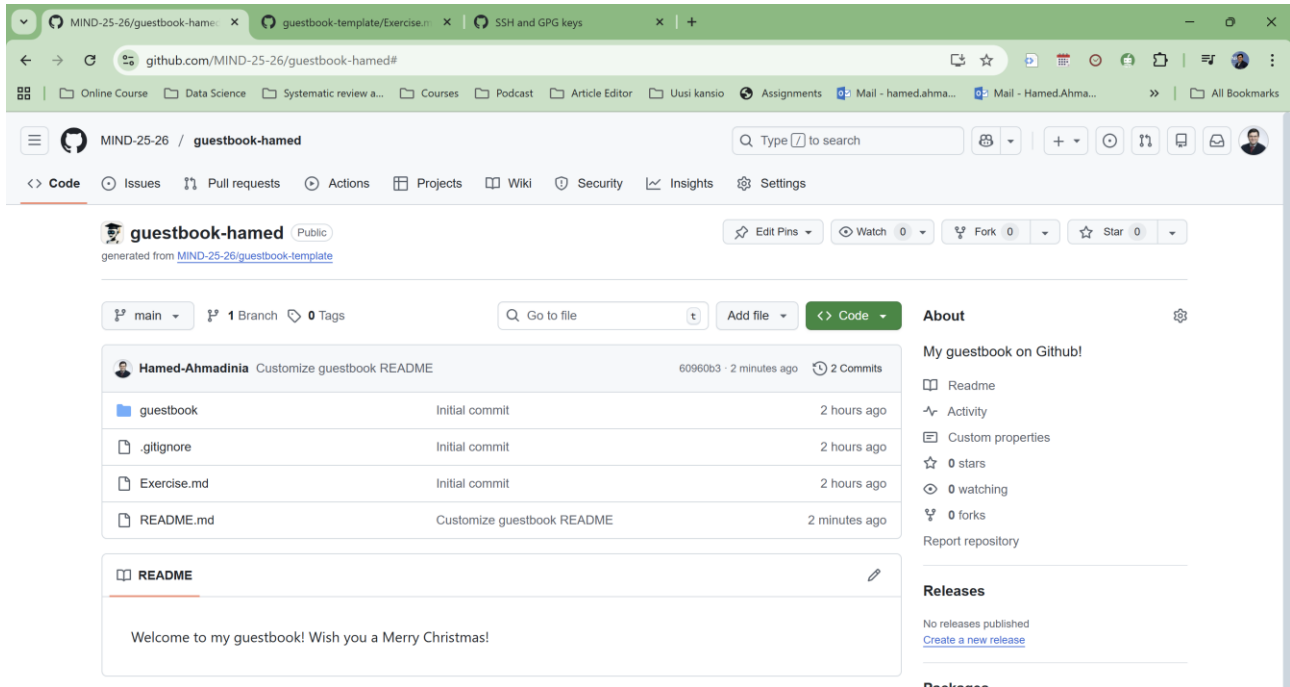
```
1 Welcome to my guestbook!
2 Wish you a Merry Christmas!
```

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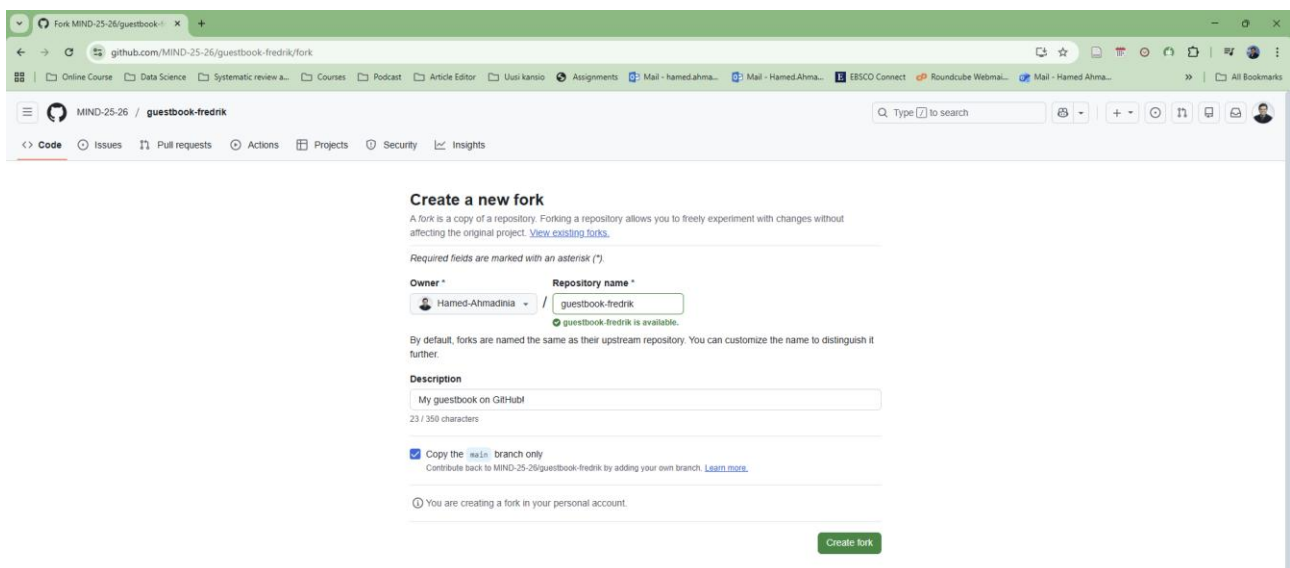
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Step 9: I verified on GitHub that my changes were successfully pushed to the remote repository, confirming that the updated README.md now displays my custom message and that the latest commit (“Customize guestbook README”) appears in the commit history.



## Part 2 – Write in someone else’s guestbook

Step 10: I forked another student’s guestbook repository into my personal GitHub account, creating my own copy of the repository so that I could make changes without affecting the original project.

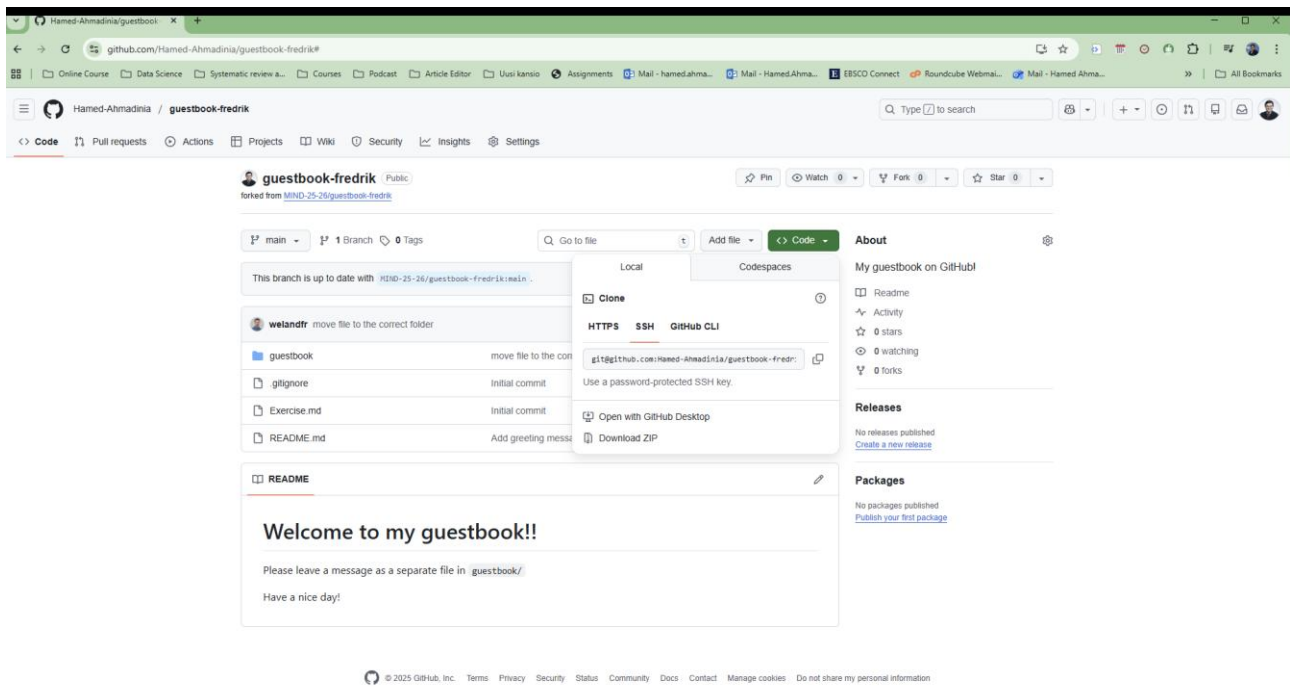


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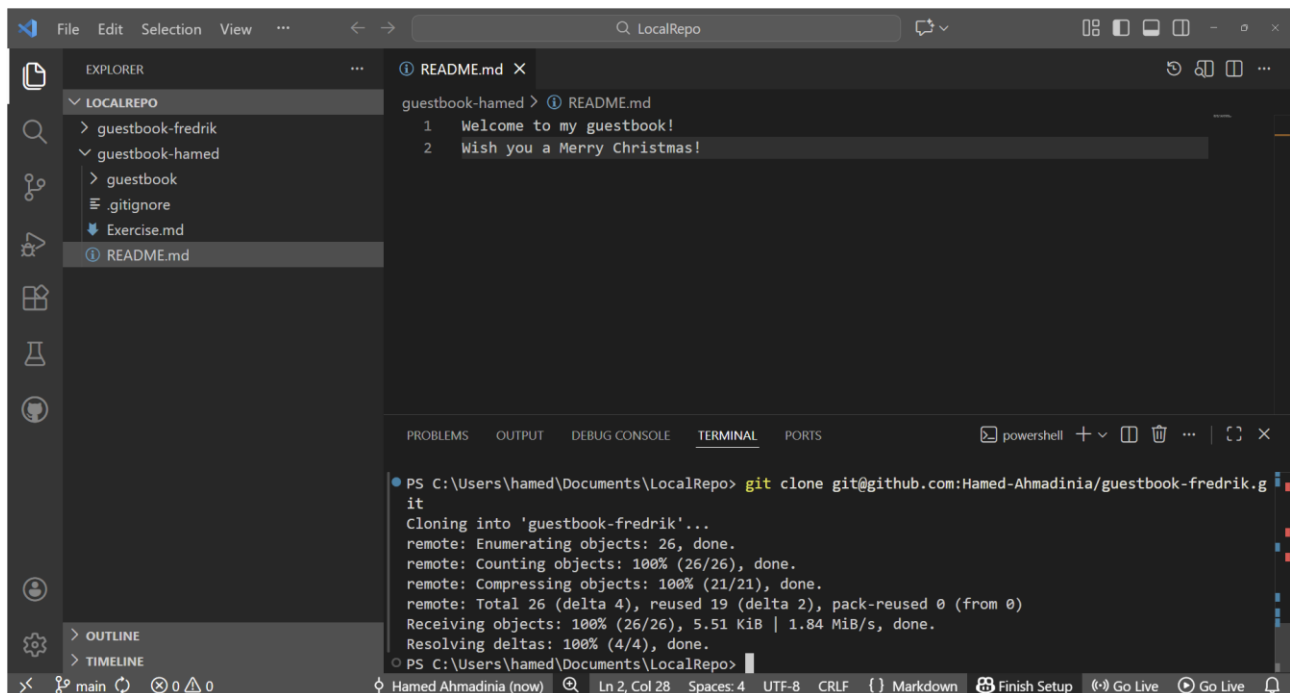
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*Step 11: I opened my forked guestbook repository on GitHub and accessed the Code menu to copy the repository URL, which I later used to clone the project to my local computer.*



*Step 12: I cloned my forked guestbook repository (guestbook-fredrik) to my local working directory using the VS Code integrated terminal, confirming that the repository was successfully copied to my computer.*



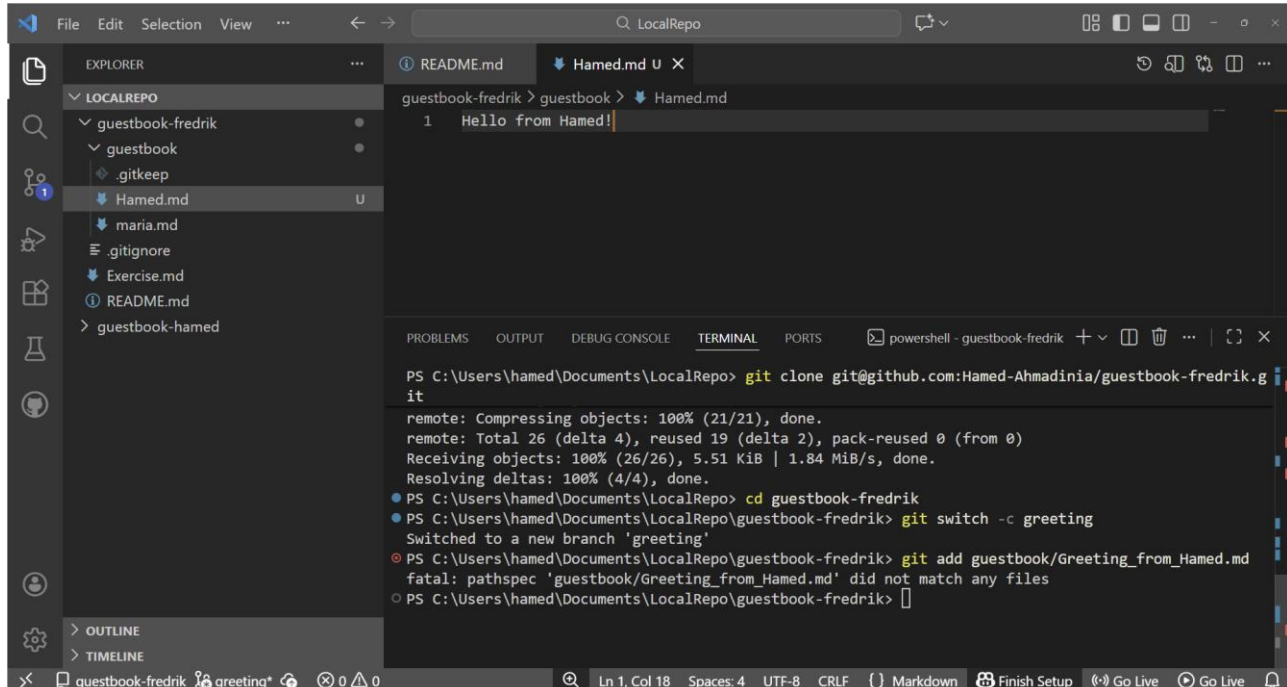


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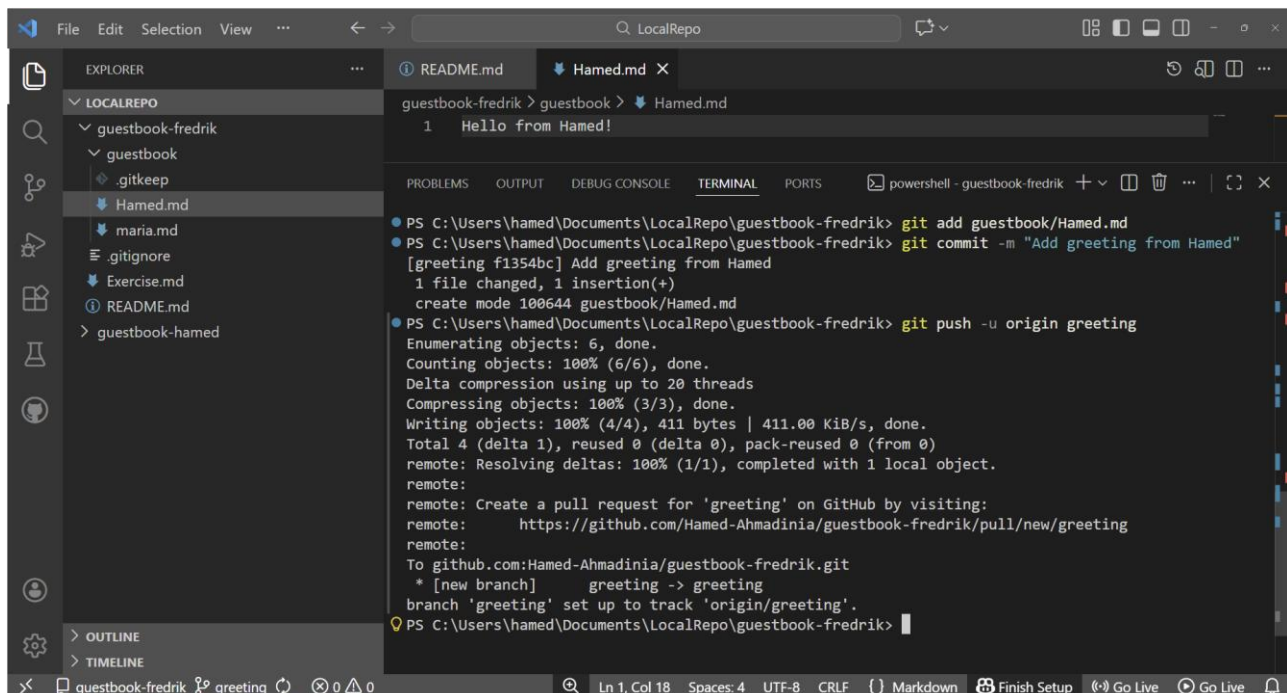
*Step 13: I created a new branch named greeting and added a new Markdown file inside the guestbook folder containing my greeting message, while also encountering and correcting a file path error during the process (inside the folder of guestbook-fredrik):*



The screenshot shows the Visual Studio Code interface with the Explorer view on the left, the Editor view in the center, and the Terminal view at the bottom. The Explorer view shows the file structure of the 'LOCALREPO' with a subfolder 'guestbook-fredrik' containing a 'guestbook' folder. The 'guestbook' folder contains files like '.gitkeep', 'Hamed.md', 'maria.md', '.gitignore', 'Exercise.md', and 'README.md'. The Editor view shows the 'Hamed.md' file with the content 'Hello from Hamed!'. The Terminal view shows the following commands and output:

```
PS C:\Users\hamed\Documents\LocalRepo> git clone git@github.com:Hamed-Ahmadinia/guestbook-fredrik.git
remote: Compressing objects: 100% (21/21), done.
remote: Total 26 (delta 4), reused 19 (delta 2), pack-reused 0 (from 0)
Receiving objects: 100% (26/26), 5.51 KiB | 1.84 MiB/s, done.
Resolving deltas: 100% (4/4), done.
PS C:\Users\hamed\Documents\LocalRepo> cd guestbook-fredrik
PS C:\Users\hamed\Documents\LocalRepo\guestbook-fredrik> git switch -c greeting
Switched to a new branch 'greeting'
PS C:\Users\hamed\Documents\LocalRepo\guestbook-fredrik> git add guestbook/Greeting_from_Hamed.md
fatal: pathspec 'guestbook/Greeting_from_Hamed.md' did not match any files
PS C:\Users\hamed\Documents\LocalRepo\guestbook-fredrik>
```

*Step 14: I staged the new greeting file, committed it with a descriptive commit message, and pushed the new greeting branch to my forked repository on GitHub, successfully preparing it for a pull request.*



The screenshot shows the Visual Studio Code interface with the Explorer view on the left, the Editor view in the center, and the Terminal view at the bottom. The Explorer view shows the file structure of the 'LOCALREPO' with a subfolder 'guestbook-fredrik' containing a 'guestbook' folder. The 'guestbook' folder contains files like '.gitkeep', 'Hamed.md', 'maria.md', '.gitignore', 'Exercise.md', and 'README.md'. The Editor view shows the 'Hamed.md' file with the content 'Hello from Hamed!'. The Terminal view shows the following commands and output:

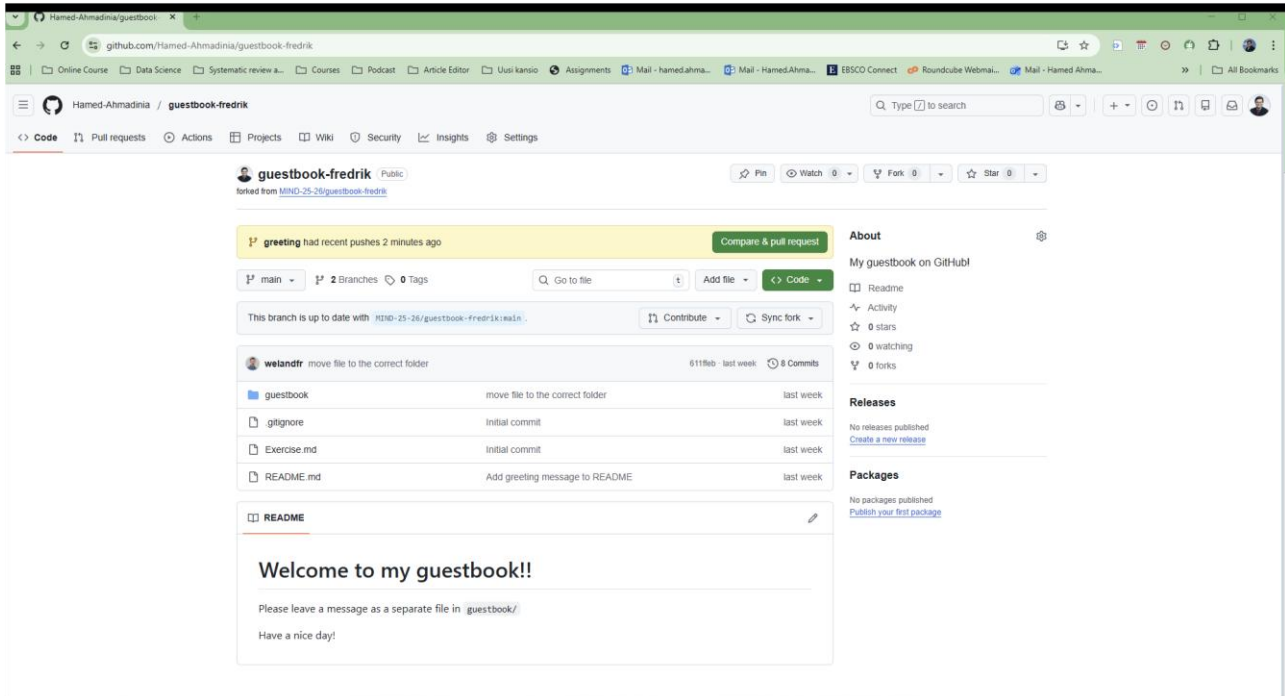
```
PS C:\Users\hamed\Documents\LocalRepo\guestbook-fredrik> git add guestbook/Hamed.md
PS C:\Users\hamed\Documents\LocalRepo\guestbook-fredrik> git commit -m "Add greeting from Hamed"
[greeting f1354bc] Add greeting from Hamed
1 file changed, 1 insertion(+)
create mode 100644 guestbook/Hamed.md
PS C:\Users\hamed\Documents\LocalRepo\guestbook-fredrik> git push -u origin greeting
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 20 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 411 bytes | 411.00 KiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'greeting' on GitHub by visiting:
remote:   https://github.com/Hamed-Ahmadinia/guestbook-fredrik/pull/new/greeting
remote:
To github.com:Hamed-Ahmadinia/guestbook-fredrik.git
 * [new branch]      greeting -> greeting
branch 'greeting' set up to track 'origin/greeting'.
PS C:\Users\hamed\Documents\LocalRepo\guestbook-fredrik>
```

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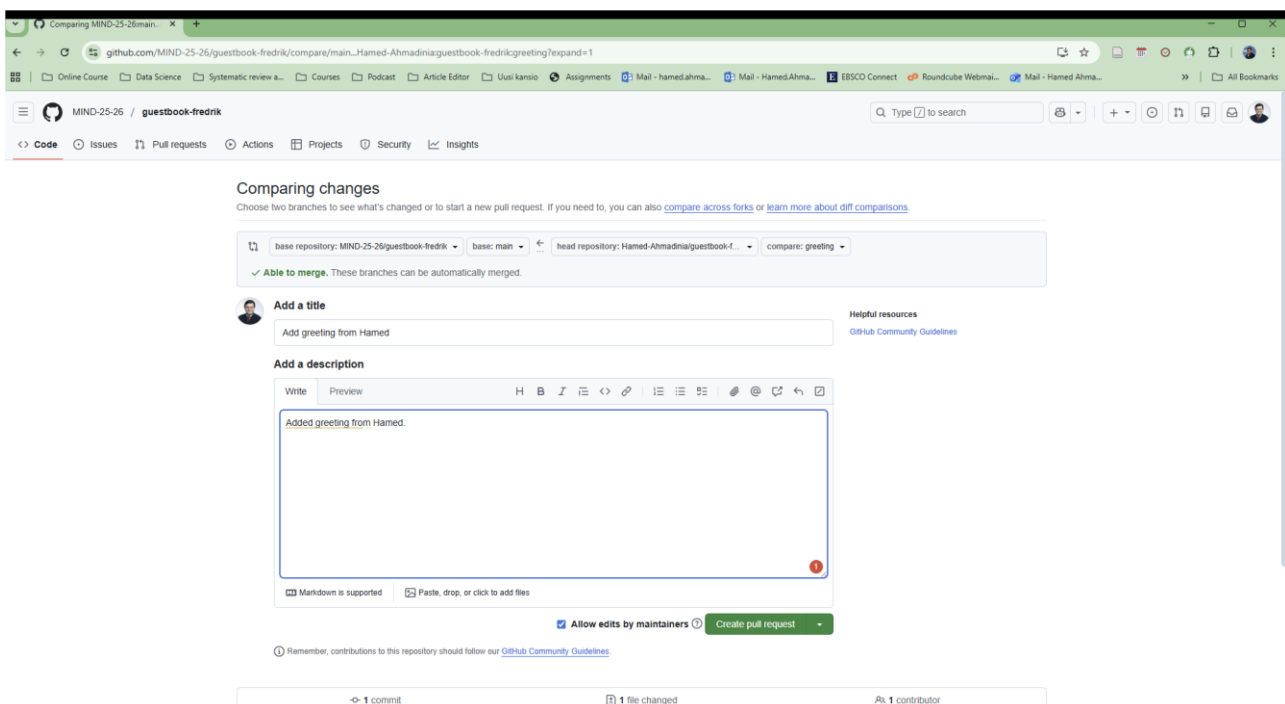
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*Step 15: After pushing my changes, I returned to my forked repository on GitHub and saw the notification banner indicating that my greeting branch had new commits and was ready to be compared and submitted as a pull request.*



*Step 16: I opened the comparison view on GitHub, verified that my greeting branch could be merged into the original repository's main branch, and added a title and short description before creating the pull request.*



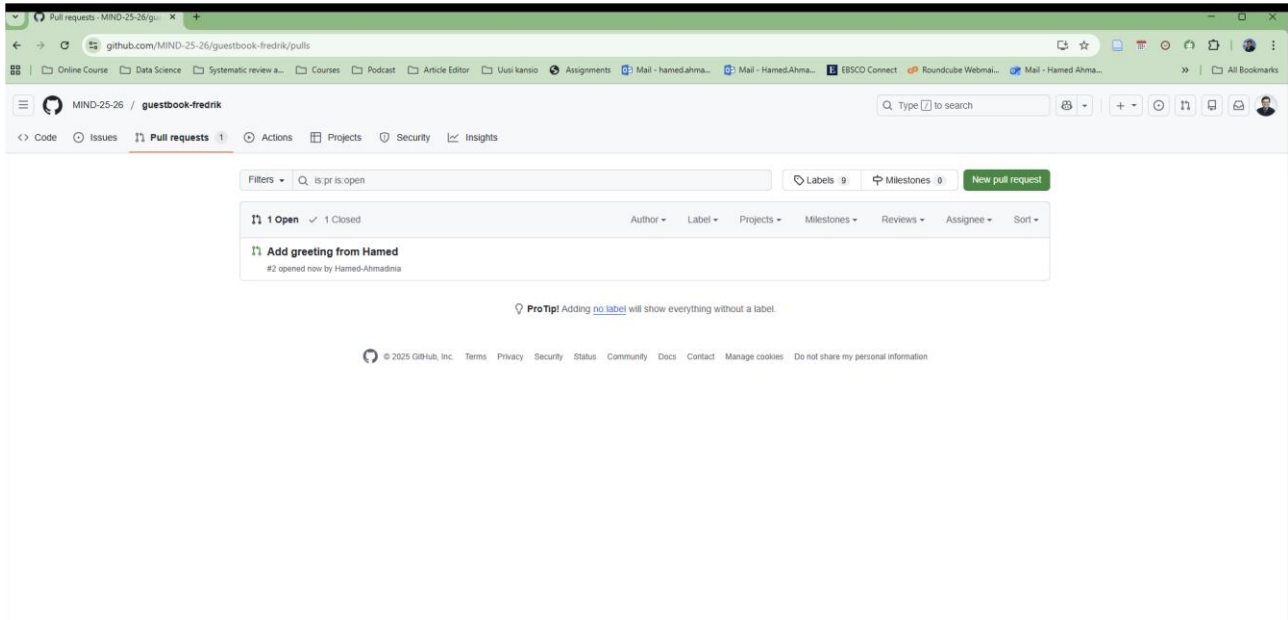


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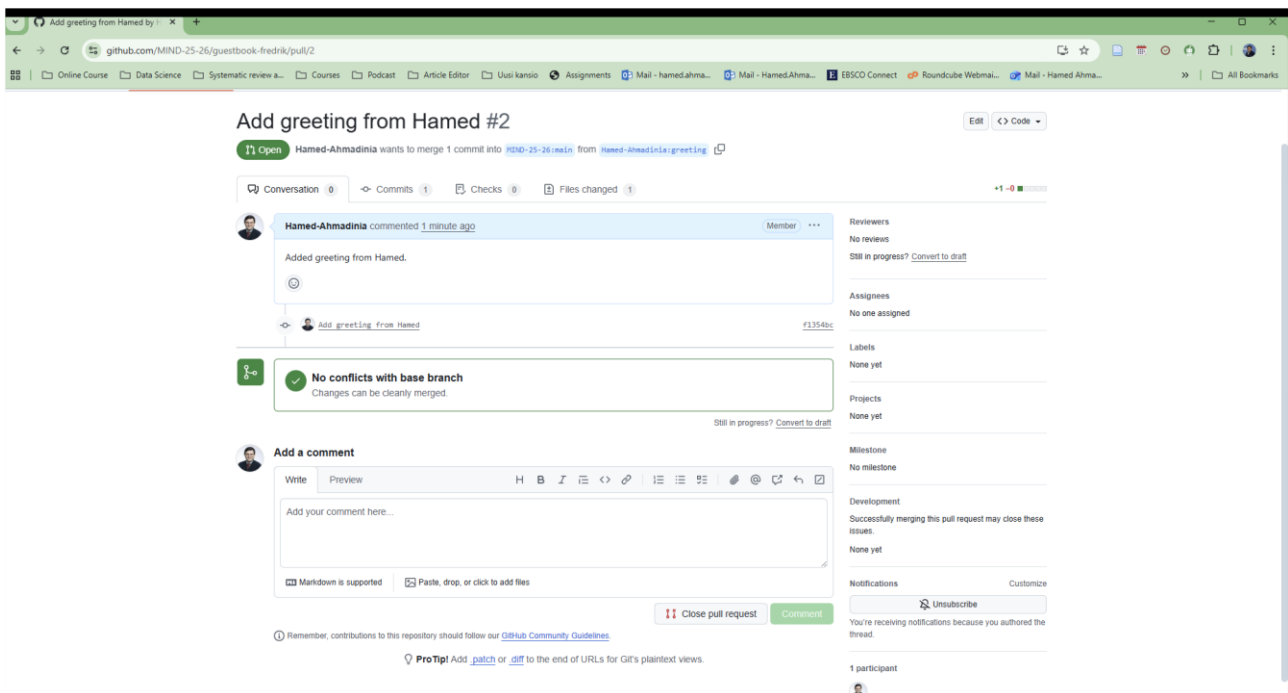
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Step 17: I viewed the Pull Requests section of the original *guestbook-fredrik* repository and confirmed that my pull request titled **“Add greeting from Hamed”** was successfully submitted and listed as an open pull request.



Step 18: I opened my pull request titled **“Add greeting from Hamed”** and reviewed its details, confirming that there were no merge conflicts and that the changes could be cleanly merged into the base repository.

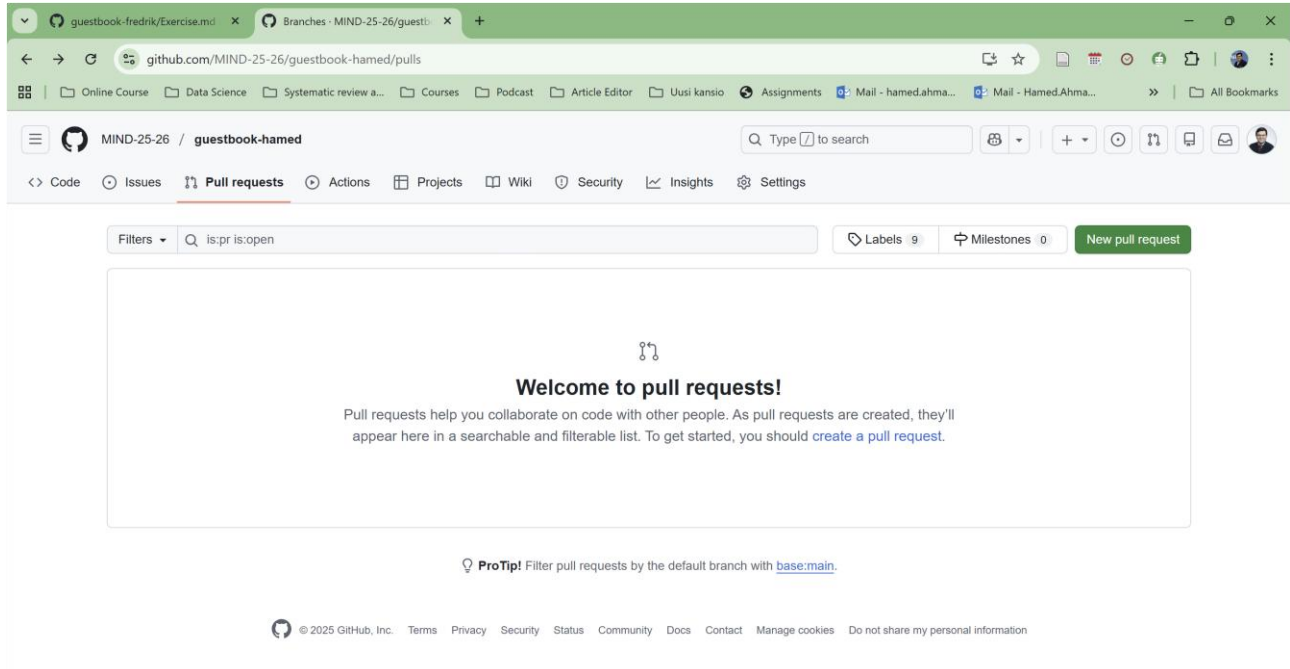


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*Step 19: I checked the Pull Requests section of my own guestbook repository to look for incoming pull requests from other students, confirming that there were currently no open requests waiting to be reviewed or merged.*



*Step 20: I verified the Pull Requests page of my own guestbook repository again, confirming that there were no open pull requests from other students to review or merge at that moment.*

